

RNAseq2tracks v2 specification

Overview

RNAseq2tracks v2 is a simplified RNA-seq preprocessing workflow intended for inexperienced users who need a one-command route from local FASTQ files to STAR alignments, gene count matrices, QC reports, and UCSC-friendly coverage tracks.

Version 2 keeps the modular structure of v1 and adds a dedicated conventional-chromosome filter before BigWig creation. This prevents non-standard contigs, alternative haplotypes, random scaffolds, decoys, and unplaced references from appearing in the browser-ready BigWig files.

Workflow design

flowchart TD

```
A[Local FASTQ folder] --> B[FastQC raw]
B --> C[Trim Galore]
C --> D[FastQC trimmed]
D --> E[STAR alignment]
E --> F[STAR GeneCounts]
F --> G[Raw counts]
G --> H[DESeq2 size factors]
H --> I[Normalized counts]
E --> J[bedtools genomecov]
J --> K[Conventional chromosome filter]
K --> L[Raw bedGraph]
K --> M[Normalized bedGraph]
M --> N[Sorted bedGraph]
N --> O[BigWig]
B --> P[MultiQC]
D --> P
E --> P
```

Version 2 changes

1. `scripts/run_coverage.sh` now filters coverage to conventional chromosomes before BigWig conversion.
2. `config/config.sh` now includes:
 - `REGULAR_CHROMS_ONLY="true"`
 - `CHROMOSOME_NAMING="ucsc"`
3. The README now has a GitHub-front-page workflow schematic and layout modeled after the style of the companion `fastq2tracks` repository.

4. The repository now includes `.gitignore`, `LICENSE`, `CITATION.cff`, `examples/`, `tests/`, and a Mermaid schematic file.

Conventional chromosome rules

When `REGULAR_CHROMS_ONLY="true"`, the coverage script keeps only these chromosome names:

Species Naming Kept chromosomes

human ucsc chr1-chr22, chrX, chrY, chrM

mouse ucsc chr1-chr19, chrX, chrY, chrM

human ensembl 1-22, X, Y, MT

mouse ensembl 1-19, X, Y, MT

The default is UCSC naming because the goal is to create browser-friendly BigWig files.

Input

The workflow starts from a local FASTQ directory. It supports paired-end or single-end mode through the fifth argument of the master script:

```
./RNAfastq2tracks.sh <input_fastq_dir> <output_dir> <max_jobs> <human|mouse>
```

Expected paired-end naming example:

K0_12_1_1__ERR14875937_1.fq.gz

K0_12_1_2__ERR14875937_2.fq.gz

Outputs

The workflow creates:

- FastQC reports before trimming
- Trim Galore outputs
- FastQC reports after trimming
- STAR sorted BAM files
- STAR ReadsPerGene.out.tab files
- raw count matrix
- DESeq2 size factors
- normalized count matrix
- raw bedGraph files after conventional-chromosome filtering
- normalized bedGraph files after conventional-chromosome filtering
- normalized BigWig files built only from conventional chromosomes

- MultiQC report
- UCSC track lines

Notes on BigWig compatibility

The BigWig files created by v2 are safer for UCSC-style use because non-standard chromosomes are filtered out before conversion. Browser compatibility still requires that your `CHROM_SIZES_HUMAN` or `CHROM_SIZES_MOUSE` file uses the same naming convention as the BAM files. If your STAR index was built with UCSC-style names, use a UCSC-style `chrom.sizes` file. If your STAR index was built with Ensembl-style names, set `CHROMOSOME_NAMING="ensembl"` and use a matching chromosome size file.

Limitations

- The workflow does not download data from SRA/GEO.
- The workflow does not build STAR indexes.
- The workflow does not perform full differential expression testing, although it prepares count matrices and normalization outputs.
- The current coverage script creates one BigWig per sample from the whole BAM. It does not split reads into separate plus/minus strand BigWig files unless further modified.
- Correct interpretation of strandedness still depends on correct sample metadata.

Recommendation

Use v2 instead of v1 when you intend to upload BigWig tracks to UCSC or another browser that expects conventional chromosome names. Keep `REGULAR_CHROMS_ONLY="true"` unless you explicitly need alternative contigs or scaffold-level coverage.